

Vehicle & Pedestrian Tracking in Surveillance Video Sequences

End-to-End Multi-Object Tracking Pipeline Featuring Fine-Tuned YOLOv8n, Kalman Filter Motion Prediction, and ByteTrack Low-Confidence Association.

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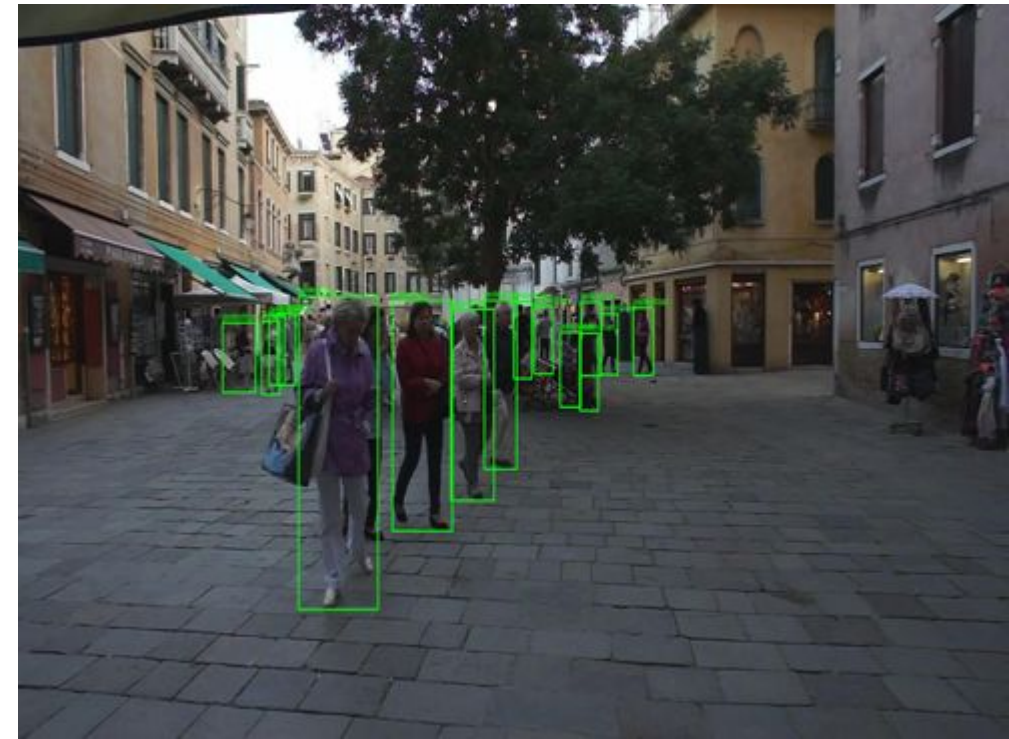
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The Problem

Given a continuous video stream from a fixed surveillance camera, automatically detect every pedestrian in each frame and assign them a persistent identity number , so the same person keeps the same ID even as they walk, occlude each other, or leave and re-enter the scene.

Two Stages Required

- ① Detection : find pedestrians in each frame, draw a bounding box
- ② Tracking : link boxes across frames with stable IDs



MOT16 Benchmark Dataset

The project uses the standard computer vision research benchmark **MOT16**. The dataset consists of human-annotated ground-truth (GT) sequences captured across varied public environments.

Sequence ID	Resolution	FPS	Frames	Environment & Camera View	Key Challenges
MOT16-02	1920 × 1080	30	600	Venice Plaza (Italy), Static Elevated	Moderate density, frequent crossing
MOT16-04	1920 × 1080	30	1,050	ETH Bahnhof (Zurich), Low Angle	High pedestrian density, occlusion
MOT16-05	640 × 480	14	837	Venice Pedestrian Street, Low FPS	Camera jitter, lower image resolution
MOT16-09	1920 × 1080	30	525	ETH Zurich Sunny Street Scene	High illumination variation, shadows
MOT16-11	1920 × 1080	30	900	Indoor Shopping Mall (PETS09)	Tile reflections, variable viewpoints
MOT16-13	1920 × 1080	25	750	Town Centre (UK Street), High Traffic	Vehicular interaction, background clutter

Total Volume: ► 5,316 frames total across all sequences

Coordinate Conversion Math

MOT16 to YOLO Format Transformation

MOT16 stores annotations as absolute pixel top-left corners: [left, top, width, height]. YOLOv8 requires normalized box centers: [class, cx, cy, w, h].

$$c_x = \frac{\text{left} + \frac{w}{2}}{W_{\text{img}}}, \quad c_y = \frac{\text{top} + \frac{h}{2}}{H_{\text{img}}}$$

$$w_{\text{norm}} = \frac{w}{W_{\text{img}}}, \quad h_{\text{norm}} = \frac{h}{H_{\text{img}}}$$

Executed by preprocess.py across all 7 video sequences to produce 5,316 individual annotation files.

Split Strategy & Label Integrity

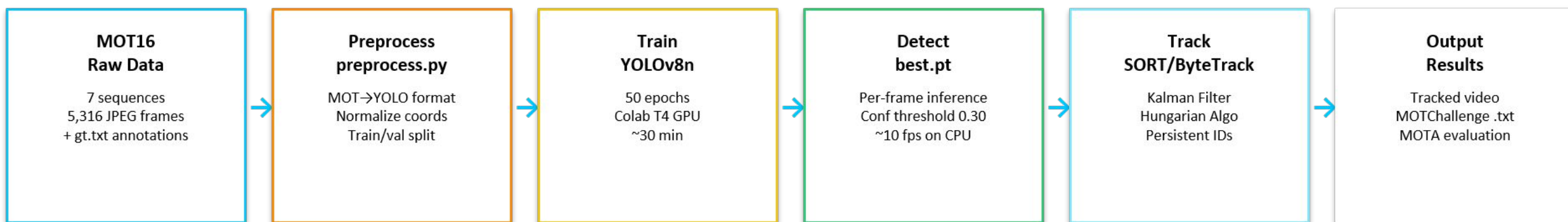
Temporal Partitioning (No Data Leakage): Consecutive video frames possess near-identical spatial representations. Standard random splitting causes severe validation leakage.

- **Train Split (80%):** First 4,254 frames (e.g., frames 1–480 of MOT16–02).
- **Val Split (20%):** Last 1,062 frames (e.g., frames 481–600 of MOT16–02).

Verification Script (verify_labels.py): Performed mathematical inversion back to pixel space and overlay bounding boxes on raw RGB frames to verify spatial precision.

Integrity Check: Zero label offset errors detected across all normalized coordinates.

End-to-End System Pipeline



YOLOv8n Detector Training

Model Architecture & Hyperparameters

Selected **YOLOv8n (Nano)** to balance high spatial accuracy with real-time edge processing constraints.

- **Backbone:** Modified CSPDarknet53 with C2f feature extraction modules.
- **Neck:** Path Aggregation Network (PAN-FPN) for multi-scale feature pyramids.
- **Head:** Anchor-free decoupled head separating bounding box regression and classification.
- **Model Scale:** 3.2 Million parameters (5.9 MB weight file best.pt).
- **Hardware Execution:** Google Colab Tesla T4 GPU (~30 min wall-clock time).

Transfer Learning & Loss Formulations

Initialized with COCO pretrained weights and fine-tuned on the MOT16 pedestrian domain.

Loss Objective Functions:

- **Complete IoU (CIoU) Loss:** Penalizes overlap, distance, and aspect ratio.
- **Distribution Focal Loss (DFL):** Regresses sub-pixel box boundaries.
- **Varifocal / BCE Loss:** Balances asymmetric positive/negative classes.

Input resolution standardized to 640 × 640 px with batch size 16 and SGD optimizer with cosine learning rate scheduling.

Loss Convergence & Accuracy

0.9058

MAP@50 (VALIDATION)

0.5915

MAP@50-95 (STRICT)

0.9237

PRECISION (PEDESTRIANS)

0.8308

RECALL (PEDESTRIANS)

Metric Target	Epoch 1	Epoch 25	Epoch 50	Total Delta	Technical Interpretation
box_loss (Regression)	1.388	0.812	0.650	-53.2%	Precise boundary alignment to ground-truth figures
cls_loss (Classification)	1.189	0.542	0.367	-69.1%	Zero confusion between pedestrian and background noise
dfl_loss (Distribution)	1.060	0.915	0.836	-21.1%	High sub-pixel confidence on sharp edge regions
Precision	0.875	0.910	0.938	+7.2%	Minimal false alarm rate on background distractors
Recall	0.728	0.784	0.809	+11.1%	Drastically reduced misses of occluded pedestrians

SORT Baseline Architecture

Kalman Filter Kinematic State Model

SORT models pedestrian dynamics using a 7-dimensional linear state vector assuming a constant velocity motion model:

$$\mathbf{x} = [u, v, s, r, \dot{u}, \dot{v}, \dot{s}]^T$$

Where u, v represent box center, s is scale (area), and r is constant aspect ratio.

State covariance matrices P and process noise Q propagate uncertainty between frame intervals.

Hungarian Association & Hyperparameters

Predicted tracklet boxes are mapped to incoming detections by solving the maximum weight bipartite matching problem via Hungarian algorithm:

$$C_{i,j} = 1 - \text{IoU}(b_i, d_j)$$

- **IoU Matching Gate:** Rejects pairs with $\text{IoU} < 0.45$
- $\text{min_hits} = 3$: Suppresses single-frame detector false alarms.
- $\text{max_age} = 3$: Destroys tracklets lost for >3 consecutive frames.

ByteTrack Two-Pass Matching

The Flaw in Conventional Trackers

Standard SORT applies a strict global detection threshold ($\text{conf} \geq 0.30$). All low scoring detections are discarded.

The Occlusion Trap:

When a target is partially occluded by another pedestrian or lamppost, the neural detector outputs a degraded confidence score (e.g., 0.22). SORT throws it away → track dies → reappearance creates a new ID.

Result: Extreme trajectory fragmentation and high identity switches (IDS).

ByteTrack Hierarchical Association

ByteTrack resolves this by utilizing **every** detection box via a two-stage matching hierarchy:

- **Stage 1 (High-Confidence):** Match detections with to existing active tracklets using Hungarian IoU. $\text{conf} \geq 0.30$
- **Stage 2 (Low-Confidence Recovery):** Take remaining unmatched tracklets and match against low-confidence detections ($0.10 \leq \text{conf} < 0.30$).

Key Result: Retains target persistence through severe occlusions without introducing false new tracks.

SORT vs ByteTrack Benchmarks

Evaluation conducted on sequence MOT16-02 (600 frames, 8,840 ground truth object instances) processed using unified YOLOv8n detector outputs.

Metric Dimension	SORT (Baseline M3)	ByteTrack (Upgrade M4)	Delta / Gain	Engineering Trade-off Analysis
Identity Switches (IDS)	98	40	-59.2%	ByteTrack maintains uninterrupted identity over long occlusions
True Positives (TP)	7,486	7,656	+170	Pass 2 recovers 170 previously missed occluded pedestrian states
False Negatives (FN)	1,354	1,184	-12.6%	Fewer missed pedestrians throughout sequence duration
False Positives (FP)	1,338	1,827	+489	Lower threshold accepts marginal noise & background reflections
MOTA Score	0.684	0.655	-0.029	MOTA penalizes FP equally to FN; higher FP slightly lowers MOTA
MOTP (Localization Error)	0.151	0.161	+0.010	Low-scoring boxes exhibit marginally looser bounding box borders

Key Finding: While SORT achieves slightly higher numerical MOTA due to conservative thresholding, ByteTrack is vastly superior for production surveillance by reducing tracking loss and identity swaps by 59.2%.

CLEAR MOT Evaluation Metrics

Multiple Object Tracking Accuracy (MOTA)

Standard CLEAR MOT benchmark metric combining false positives, false negatives, and identity switches relative to total ground-truth targets:

$$\text{MOTA} = 1 - \frac{\sum (FP_t + FN_t + IDS_t)}{\sum GT_t}$$

SORT MOT16-02

$$1 - (1338 + 1354 + 98) / 8840 = 1 - 0.3156 = 0.684$$

ByteTrack MOT16-02

$$1 - (1827 + 1184 + 40) / 8840 = 1 - 0.3451 = 0.655$$

Multiple Object Tracking Precision (MOTP)

Measures spatial overlap precision between predicted bounding boxes and matched ground-truth boxes:

$$\text{MOTP} = \frac{\sum_{t,i} (1 - \text{IoU}_{t,i})}{\sum_t c_t}$$

Where c_t is the number of true positive matches found in frame t . Lower values denote superior localization alignment.

Both models demonstrate high spatial accuracy with IoU error below 16.1% on standard 1080p surveillance video feeds.

Conclusions & Future Scope



Engineered Milestones

Successfully transformed 79,799 MOT16 labels, trained YOLOv8n to 0.906 mAP@50, and built end-to-end SORT and ByteTrack pipelines running at 10 FPS on commodity CPU.



Encountered Challenges

Severe camera motion and distant small targets (<15px) test detector limits. Trade-off between false positive acceptance and track continuity requires fine tuning.



Future Extensions

Deploy DeepSORT appearance feature extractors (Re-ID embeddings) to handle long-term re-emergence; expand model training to BDD100K for multi-class vehicular tracking.

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Department of Computer Engineering | University of Peradeniya

github.com/cepdnacl/e23-co5430-vehicle-pedestrian-tracking

✓GROUP G07 VALIDATED